Challenges in the Everyday Lives of the Employed DHH Community in India

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One-sixth of the global DHH (Deaf or Hard-of-Hearing) population resides in India, constituting over 63 million people. Still, research on the DHH population has been mostly concentrated in the Global North. In this qualitative and quantitative study, we conducted 15 interviews and surveyed 131 people from the employed DHH community in India to understand accessibility issues faced by them daily. We focus on the employed DHH community as they are critically dependent on communication in variety of settings and are usually the early adopters of assistive technology. Our work reveals that a majority of the employed DHH population face challenges at the workplace, and that banks, hospitals, family conversations, phone calls, and public transport are also inaccessible to them. Accessibility issues include the non-availability of sign language interpreters at crucial settings (such as workplace, banks, and hospitals), and workarounds such as exchanging written messages, auto-generated captions, lip-reading, and using basic gestures are inadequate. This results in an inability to participate in workplace meetings, poor interactions with colleagues and manager, difficulties in using service providers such as e-commerce or ride hailing applications, and a dependence on friends and family to navigate various everyday scenarios. Based on insights from these outlined challenges, we propose technology and policy-related recommendations to improve accessibility for the Indian employed DHH community.

CCS Concepts: • **Human-centered computing → Empirical studies in accessibility**; *Accessibility design and evaluation methods*; **User studies**.

Additional Key Words and Phrases: Accessibility, Disability, Assistive Technologies, Workplace, Empirical study, Deaf, Hard-of-Hearing

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1 INTRODUCTION

The Deaf or Hard-of-Hearing (DHH) community constitutes over 5% of the global population [4]. Due to the inaccessible nature of their environments, this community faces a variety of challenges in their everyday lives. Most of the accessibility research work that study these challenges is based in the Global North[1]. These studies explore the socioeconomic context of the lives of the DHH community, such as the role of family [34], communication challenges in educational institutions [54], and access to healthcare [46]. Prior work in resource-rich countries has also explored accessibility issues faced by DHH employees in a workplace setting—examining communication patterns, career barriers and difficulties faced by the DHH community in the Australian workplace [73], communication preferences of DHH professionals in supervisory roles in the USA [37], and a literature survey about experiences of stress and fatigue among the members of the DHH community at work in USA, Europe and Australia [71]. Other studies assume well-resourced settings and explore technological interventions to aid communication between DHH and hearing people, like custom camera

[1]Global North refers to the countries located primarily in the northern hemisphere that have historically been identified as "the West" or "first world" due to perceptions of their relative wealth, technology, and global dominance.

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setups to recognize sign language [3, 15], use of video/telecommunications relay services to make calls [8, 83], and teletypewriter technology on smartphones for emergency calls [86].

Many of these solutions and insights do not generalize to a country like India due to the large differences in cultural and economic context, access to technology, and employment opportunities. One example of this cultural difference is the nature of Indian Sign Language (ISL). Unlike American or British Sign Language, ISL is not a *single* language. It has a variety of dialects, such as Delhi Sign Language, Kolkata Sign Language, Bangalore-Chennai Sign Language, and Bombay Sign Language [1]. This diversity has impeded standardization and makes communication within the Indian DHH community hard. The structure of ISL is also exclusive—the ISL lexicon has a high level of iconicity and compounding not seen in other sign languages [33].

Other factors specific to India include accessibility-related cultural taboos (reported in work on the Indian blind population [47, 49]), limited opportunities for education in mainstream schools [47], and the lack of trained interpreters [51]. The Indian DHH population also warrants its own study due to its large size—India is home to 63 million DHH people or one-sixth of the global DHH population [28, 38]. Additionally, hearing loss is the second most common disability in India, representing 18.9% of the disabled population [66]. Despite the above, formal studies on the challenges faced and potential support systems for the DHH community in India has not received much research attention.

In this work, we seek to address and focus on a part of this gap by asking the research question: *"What are the key communication challenges faced by the working Indian DHH community today?"* We focus on communication as it is a fundamental challenge faced by the DHH community [37, 73]. Additionally, we focus on the employed DHH demography, due to: (a) the importance of employment for earning a livelihood, developing skills, and collaborating with others [55], (b) the challenging communication scenarios faced by them including at home, workplace, and commute, and (c) the potential for them to be early adopters of assistive technology due to financial independence.

The DHH community has historically experienced higher rates of unemployment and lower levels of education compared to people without hearing loss [72]. Additionally, adults with hearing loss have a much higher unemployment rate in low- and middle-income countries compared to high-income countries [5], underlining the importance of such a study. With the recent push for diversity, equity and inclusion in the workplace [20], we believe technological solutions designed to make the workplace accessible for the DHH community have higher likelihood of adoption.

In order to achieve our goal, we conducted a mixed-methods study of the employed Indian DHH community. Our qualitative study consisted of 15 virtual interviews, facilitated by a certified sign language interpreter; our quantitative study consisted of 131 responses to an online form circulated amongst employed members of the DHH community. Our findings show that DHH employees face accessibility issues in a variety of everyday settings, primarily at the workplace. Non-availability of sign language interpreters at workplace, and limitations of workaround methods (like exchanging written messages, auto-generated captions, lip-reading, and using basic sign gestures), results in their exclusion from workplace meetings and limits their interactions with colleagues and manager. Apart from workplace, we identified accessibility issues in interacting with cab drivers, e-commerce/food delivery personnel, bank officials, and doctors. As a result, our participants are often dependent on friends, family, colleagues, and strangers, which reduces their agency. Based on insights from the challenges outlined above, we propose technology and policy-related recommendations to improve accessibility for the Indian employed DHH community.

2 RELATED WORK

In this section, we examine existing work in two related areas—research on the challenges faced by the DHH community and accessibility work in the Indian context. The latter has primarily focused on the visually impaired community, along with a few initial studies on ISL.

2.1 Research on challenges faced by the DHH community

The DHH community relies mainly on assistive technology to understand speech. Simple low-technology aids such as exchanging written/text messages can work, but are much slower than spoken or signed communication, hence they are not effective for sustained conversation [40]. Several technological solutions have been proposed such as video/telephone relay services [8, 9], Internet Captioned Telephone Services [7], and Assistive Listening Devices [2]. The methods used by the DHH community to communicate with hearing people have been well studied as well [37, 42, 72], and uncover several communication-related workarounds including avoidance, lip reading, optimizing volume of speech, confirming the message, attracting the attention of the hearing person, and using simpler words and signs. Communication challenges have been studied in specific contexts like home, workplace and educational settings.

Significant work has been done in the context of education given its importance in shaping people's lives, such as the experiences of DHH students in educational institutions [54, 67], hearing students' perspectives on the inclusion of DHH students [25, 44], methods to teach and assess DHH students [23, 24, 62, 64], and reading experiences with assistive tools such as automatic text simplification [13, 50, 60]. These studies have found that DHH students may not always disclose their needs and often settle for sub-par accommodations [22, 54]. They also dismantle misconceptions that technology can remove all access barriers or that technology built for one individual can generalize to others [54].

Another context that has been studied extensively is the home. The home plays a sensitive role in shaping experiences of DHH children, since more than 90% of deaf children are born to hearing parents [6]. Hearing parents of DHH children have to make several decisions regarding communication and language choices on behalf of their children. Research on the factors that influence this decision-making [31, 45] reveal that parents learn to view hearing loss in one of two ways: the *sociocultural* view of deafness aligning with the use of sign language, or the *audiological* view aligning with the use of the family's spoken language. The efficacy of these methods depends on the characteristics of the DHH child. However, it has been observed that parents often choose a method based on inadequate information obtained from professionals and/or the priorities that they have for their children [85].

Lastly, an analysis of the workplace context is crucial due to its potential to support accessibility interventions. Several studies chronicle the current experiences of the DHH community in the workplace and propose and evaluate approaches to reduce the challenges faced by the community [35, 37, 63, 65, 70, 71, 73]. Quantitative findings from a survey-based study indicate that the most difficult workplace situations were meetings, training activities, and work-related social functions [73]. There exist multiple accommodations to address these challenges, such as better lighting and furniture rearrangement to facilitate seeing other people better, access to sign language interpreters, and Automatic Speech Recognition (ASR) for live captioning during meetings [63]. Apart from workplace provided accommodations, it is apparent that certain personal attributes like persistence and self-advocacy skills, as well as the presence of supportive and helpful co-workers are important to facilitate job satisfaction and success among the DHH community [32]. There has also been research on the awareness of the Americans with Disabilities Act (ADA)[2] [81] among employers, and the perspective of hearing managers of Deaf workers [82].

[2]ADA is a comprehensive civil rights law in USA prohibiting discrimination based on disability in employment, state and local government programs, public accommodations, commercial facilities, transportation and telecommunications.

2.2 Accessibility Research in India

Most of the above cited work on the DHH community has been localized to the Global North, where both the resources and awareness to address these challenges are relatively higher compared to low- and middle-income countries. In the wider space of accessibility research, significant work has been done over the past decade on the visually impaired community in India. This body of research has studied various aspects of the life of people with vision impairment, such as navigation [52, 53], education [48, 49], technology adoption, and has proposed accessible solutions [56, 74, 84]. Beyond the visually impaired community, there has also been research on developmental disabilities such as autism [12, 16, 76], cerebral palsy [78], and speech and motor impairment [30]. There is limited work on the DHH community in India. Existing work has primarily focused on educating DHH children [66, 87], methodologies to teach English to DHH students [36, 69], and teachers' perspectives on the education of DHH children [68]. We did not find any existing research that studies the broader set of challenges faced by the DHH community in India or that specifically focuses on employed DHH.

More recently, researchers have studied the Indian Sign Language (ISL) [79, 80]. These studies focus on the diversity of ISL and its unique characteristics that differ from other sign languages as well as the many spoken languages of India. They examine the several regional dialects of ISL used across India [1], and its high iconicity and use of compound signs [33]. Another crucial feature of ISL that has been examined is its variation with class differences. Members of the educated middle-class DHH community use a unified and relatively standardized sign language, while the rural DHH have no exposure to this urban form. Instead, the rural DHH community use organically evolved *home-sign systems* [75], specialized to the socio-linguistic context of their communities. To further research on ISL and promote its standardisation and usage, the Government of India established the Indian Sign Language Research and Training Centre in 2015. Moreover, as part of a new National Education Policy announced in July 2021, the Indian government launched an ISL dictionary of 10,000 words and suggested introducing ISL in the school curriculum to improve accessibility and create awareness [57].

In summary, the challenges of the DHH community have mainly been studied in the Global North. However, many of these insights and technological solutions do not directly generalize to India due to multiple reasons like lower access to resources and education, lack of accommodations for DHH employees, and distinctive features of ISL.

3 METHOD

To understand the communication challenges faced by the employed DHH community in India, we circulated an online survey amongst employed DHH, followed by semi-structured video interviews. Both the survey and the interviews contained questions about accessibility issues faced on a day-to-day basis by this demography. The study was approved by the Institutional Review Board at <Anonymous Institute>. In this section, we present our survey and interview methodologies, along with our data collection and analyses techniques.

3.1 Survey

Our survey consisted of 18 questions—12 multiple-choice questions and 6 open-ended questions—and was administered via Google Forms. The survey was organised in three sections. The first section consisted of demography questions (such as gender, age, occupation, and level of hearing disability). The second section consisted of questions related to communication challenges and workarounds (such as "*Where do you find it difficult to communicate with hearing people?*" and "*Which apps do you use to talk to your family and friends?*"). The final section focused on the workplace

context (such as "*Does your workplace have a sign language interpreter?*", "*How often do you have meetings at work?*", and "*How do you communicate with your coworkers?*"). The final, optional question asked survey participants to provide their email and phone number in case they wanted to participate in a follow-up interview. No Personally Identifiable Information (PII) was collected unless the participants volunteered to participate in the interviews.

The survey was administered in English. All the survey questions were short, simple, and unambiguous to make them accessible to participants with low English proficiency. The survey was piloted with two researchers and a certified ISL interpreter who provided feedback on question framing. At the start of the survey, its purpose was explained in English and ISL. The survey form was distributed by a certified ISL interpreter, an NGO working for the DHH community and a philanthropic organization supporting work on the Deaf within their networks using WhatsApp and email. Furthermore, we asked the survey participants to share the survey within their WhatsApp groups. In order to be included in the survey, participants needed to be Deaf or Hard-of-Hearing, currently living in India, and employed (either currently or in the recent past).

In total, we received 131 valid responses (107 male, 24 female, age=27.9±5.9 years) over 30 days spanning Jan–Feb 2021. All the survey respondents self-attested as DHH, out of which 42.2% of respondents have profound hearing loss (*i.e.*, complete deafness), 25.9% have moderate hearing loss, and 31.9% have mild hearing loss. Our participants were educated, with 24.4% having graduate degrees, 49.6% having undergraduate degrees, and the rest having a high school diploma. At the time of answering the survey, 65.8% of survey participants worked from home, 22.2% worked solely from the office, and 12.0% worked in a hybrid model.

3.2 Interview

We conducted semi-structured interviews after the survey during Feb–Aug 2021 period. Interview participants were recruited from two sources–survey participants who volunteered to participate in the interview, and recruitment messages shared by a certified ISL interpreter within her network using WhatsApp and email. In the interviews, we asked participants about their sign language education and proficiency, how they communicate with hearing people (family, friends, coworkers, and strangers), and accessibility challenges and workarounds in various settings (like home and the workplace). We also asked them to recall a recent accessibility-related incident and the approach they used to handle that situation. At the end of the interview, participants were invited to share open comments and express any concerns.

All interviews were conducted remotely by the first two authors using the Google Meet video conferencing tool in the presence of a certified ISL interpreter (female, 27 years old). All the calls were video-recorded with the consent of the participants. Participants were informed that the data would only be used for research purposes. The interviews lasted 30–75 minutes. The authors and the ISL interpreter interacted in English, while the ISL interpreter and the participant communicated in ISL. The interviews were transcribed soon after they were conducted, and we use the exact translation provided by the interpreter when quoting participants. Participants were paid 500 INR for participation.

In total, we interviewed 15 participants (8 male, 7 female, age=21–34 years). All participants (except one) were currently employed. One participant was currently unemployed, but was employed in the last 6 months. All participants were from urban areas of India. The interview participant demographics are available in Table 1.

3.3 Data Analysis

We conducted a mixed-methods analysis to systematically analyze the collected data: quantitative analysis of surveys and grounded theory analysis of interviews. While we draw on all data sources in order to arrive at the findings,

Table 1. Demographic details of the interview participants

ID	Age	Sex	Hearing Loss Level	Location	Highest Education Degree	Sector	Occupation
P1	26	M	Mild	Hyderabad	Under-graduate	IT/BPO	Movie editor
P2	26	F	Moderate	Coimbatore	Under-graduate	IT/BPO	Security Guard
P3	37	M	Moderate	Bombay	High School	Education	Teacher
P4	34	M	Moderate	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P5	25	M	Profound	Theni	Post-graduate	Retail	Data Entry Operator
P6	25	M	Profound	Coimbatore	Under-graduate	E-commerce	Warehouse Assistant
P7	29	M	Profound	Coimbatore	Post-graduate	IT/BPO	Process Executive
P8	21	F	Profound	Hyderabad	Under-graduate	NGO	Video Creator
P9	25	F	Profound	Hyderabad	Post-graduate	IT/BPO	Customer Support
P10	24	F	Profound	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P11	32	F	Profound	Delhi	Under-graduate	Education	Teacher
P12	33	F	Profound	Hyderabad	Under-graduate	IT/BPO	Expenditure Auditor
P13	26	M	Moderate	Trivandrum	Under-graduate	Education	Teacher
P14	25	M	Moderate	Trivandrum	Under-graduate	IT/BPO	Software Developer
P15	21	F	Profound	Hyderabad	Under-graduate	NA	Currently Unemployed

we quantitatively analyzed survey responses in isolation to understand respondents' occupation, context-specific communication challenges, and technology use. We used grounded theory [39] to analyze interviews. We subjected our data to open coding and rigorously categorized our codes to examine communication challenges and workarounds by the employed DHHs in India. All authors regularly participated in the coding process and iterated upon the codes until consensus was reached. Over the course of analysis, they interacted over multiple days to: (1) discuss coding plans, (2) develop preliminary codebook, (3) review the codebook and refine/edit codes, and (4) finalize categories and themes. The first-level codes were specific, such as "communication with family", "lip-reading", and "challenges attending phone calls". After several rounds of iteration, the codes were condensed into high-level themes, such as "ISL education", "workplace-related challenges", and "communication workarounds". Several other approaches including multi-level code reviews, peer debriefing, and member checks were used to improve the credibility and validity of the coding process. Please note that we refer to survey respondents as 'respondents' and interview participants as 'participants' in the rest of the paper.

4 FINDINGS

In this section, we discuss findings from our survey and interviews. We first describe various accessibility challenges faced by the employed DHH community in the workplace. We then describe accessibility challenges faced by them in other everyday scenarios. Along with challenges, we also present workarounds. Lastly, we provide insights from our study that might explain the core reasons behind the identified challenges.

4.1 Work

We asked survey respondents—"*Where do you find it difficult to communicate with hearing people? (Select all that apply)*" The results show that a majority of our respondents (53.4%) experience communication challenges at Work (Figure 1). Hence, we focused on workplace-related challenges in our interviews.

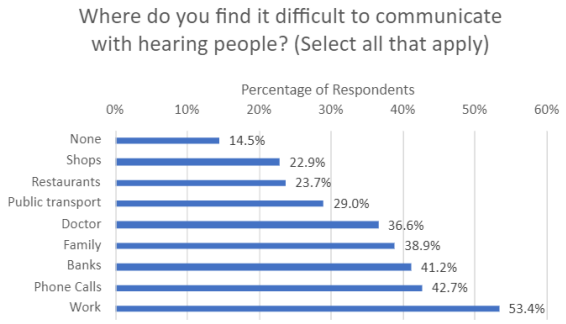

Fig. 1. Survey results for settings where our survey respondents find it difficult to communicate

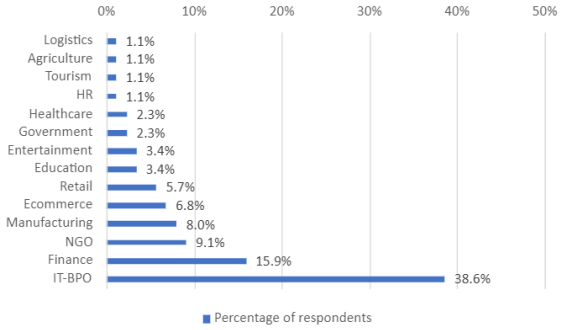

Fig. 2. Sectors where our survey respondents worked

4.1.1 Job Descriptions and Preparation. Our survey respondents were working at 53 different organizations, which we mapped to 14 sectors (Figure 2). We found that the IT-BPO (Information Technology-Business Process Outsourcing) and Finance sectors are the most common sectors employing 38.6% and 15.9% of our survey respondents, respectively. Our interview participants worked in 6 of these sectors, with a majority of them (8) working in the IT-BPO sector. Respondents of the IT-BPO sector mainly worked jobs such as software engineers (13.7%), expense auditors (13.0%), and data entry operators (6.1%). P4 described his job as an expense auditor in an IT-BPO company:

> "There is an app called Concur. We approve bills there. Bills come from third party, and we see if the amount being claimed matches the amount on the bill. If the amount is not matching, we have to manually delete it and enter the amount printed on the bill. Then we get to the next bill... We have to check each bill within 10 seconds, otherwise our reports per hour goes down. We have to reach [at least] 97% of the weekly target... We have to login for 8 hours per day, with 1 hour of break." – *P4*

We found such job descriptions to be typical. They were designed to minimise communication with other colleagues and/or customers, and required minimal English literacy. In spite of being a white-collar technical job, it was menial requiring minimal expertise. Moreover, we found 99.2% of survey respondents and all our interview participants worked in individual contributor roles, mainly because such roles require "*no communication with colleagues*". P6 stated that he was *"unable to grow"* in leadership roles in his organization due to communication challenges. As an exception, P8 did not face communication challenges in workplace, as she worked for an NGO consisting mainly of DHH employees.

Lastly, despite already having a college degree, 10 interview participants had to complete additional training courses—like English writing, typing on keyboard, using Microsoft Excel, and video editing—to get employed. Our participants found these courses to be valuable, as "*it taught* [them] *the in-demand skills*" and connected them with prospective employers, thus significantly increasing their likelihood of getting employed.

4.1.2 Access to Interpreters. 49.6% of survey respondents had access to interpreters at work, while another 8.5% had interpreters present only during major events. The rest of them (41.9%) did not have access to any interpreter at their respective workplaces. However, 76.3% of survey respondents said that they used sign language at work, indicating that sign language is preferred even if interpreters are not present at the workplace. Six of our interview participants stated that their office lacked an interpreter, but four of them still used ISL at their workplace. This was made possible using interesting workarounds, such as:

> "We don't have any interpreter. Initially we faced difficulty in talking to the supervisor but later we started teaching them sign language. We started with alphabets and a few words. Now they can sign! There is also another deaf girl who can talk, which makes things easier." – *P2*

4.1.3 Meetings. Meetings are a crucial component of office life, with employees across organizations attending meetings for an average 6 hours/week [61]. 30.2% of our survey respondents reported having meetings everyday, with another 34.1% having meetings at least once a week, 16.7% having 1-2 meetings per month, and 19.0% having no meetings. Among our interview participants, these meetings comprised of weekly performance discussions for each employee, rewards for good performance, and/or training for future assignments. When asked the question "*Are you able to actively participate in meetings?*", 53.7% of survey respondents responded with 'yes', 24.8% responded 'sometimes', and the rest 21.5% stated 'no'. To understand these challenges better, we asked our interview participants about their meetings-related experiences.

We found participants with access to interpreters at workplace are more successful at participating in meetings. In the organizations of P4, P10 and 12, the interpreter jointly leads the team with another manager. Thus, the interpreter has both work context and is comfortable with ISL. In case of P9's organization, the interpreter does not have a management role; the interpreter conducts a separate follow-up meeting with the DHH employees after a meeting concludes, to "*reiterate everything that was discussed*". Participants (like P5) who did not have access to interpreters, reported that he "*mostly skip meetings*" and rely on meeting minutes prepared by the team leader. While this workaround allowed DHH participants to be updated, it precluded them from meaningfully contributing to such discussions.

A few participants use lip reading to follow meeting discussions. For instance, P7 uses hearing aids to attend meetings. Having learnt lip reading and received speech therapy during his childhood, P7 can 'listen' and respond to others, as conversations happen during a meeting. However, this approach could get frustrating, as P7 described:

> "I feel bad when others are speaking fast and I am unable to tell them to slow down. If someone is talking to me fast and I am unable to understand then I ask someone else to explain what that person is saying. That person usually summarizes it instead of telling me the whole story and treats me like a baby... I don't like that. I sometimes turn off my video and sit because I don't understand what people are saying." – *P7*

Five participants mentioned trying the auto-generated captions feature offered by video conferencing platforms (like Google Meet). While they found the feature useful, they stated two reasons that prevented them from using live captions regularly. First, participants struggled to read and understand captions when someone spoke fast: "*If a person is speaking too fast... only 50% of the time I can understand, rest of it I miss.*" (P1). Second, the caption generator often make errors in understanding Indian accents and pronunciations, especially for Indian names.

4.1.4 Conversations with managers. Communication with one's manager is vital to success at work. Participants preferred organizations with interpreters acting as the team lead for DHH employees. Although a few such participants had another reporting manager, their single point of contact was still the (team lead) interpreter, who resolved their doubts and raised concerns with the reporting manager, as and when required. Communication between DHH employees and hearing managers is challenging in offices without interpreters. The default mode of communication is exchanging written messages, on a piece of paper or over email/WhatsApp. While written messages enable limited communication, it has two drawbacks. First, it is slow and becomes impractical for long, two-way conversations. Second, participants were concerned that writing about an issue "*appears excessively formal*" and may be taken out of context as a complaint. E.g., P6 described his hesitation with writing an email requesting promotion:

> "Initially, my [DHH] seniors helped me understand and do my role. Now I want to go to the next level... I don't like doing the same work for a long time. But [my] manager is not able to understand what I'm trying to say. If I have to write and ask about the promotion, they [the management team] will be asking questions... I am afraid that they will give me lecture on it, so its better to not talk about it." – *P6*

Two participants stated that they took the help of their hearing colleagues often to communicate with their manager. They were more comfortable in having a written exchange with their colleagues, because the colleagues were not as busy as their managers, hence our participants could have a relaxed informal conversation with them. For instance, P14, who is the sole deaf engineer in his organization, stated:

> "When I need to have some conversation with my manager, I will first talk to a coworker. Then he/she will come with me to explain to the manager. I used to write [on paper] and explain to coworker for long conversations. Now I use WhatsApp to communicate [with the coworker]... Conversations are usually about work-related issues. I never spoke about my salary with my manager." – *P14*

Other participants like P6 reported having very limited conversations with his manager. P6 works at an e-commerce warehouse that does not employ an interpreter. His job is to pack and seal products into boxes before they get shipped to customers. Due to his blue-collar work profile, he finds it difficult to communicate even over WhatsApp/email.

> "Before I joined [the team], there were Deaf people working already. They taught the manager sign language, but not much. I cannot talk about my job role to my manager. I can only say hi/bye to him. He [my manager] only understands conversations like 'did you have your tea?'." – *P6*

4.1.5 Conversations with colleagues. All participants reported that the frequency and quality of interactions with their hearing colleagues were poor. E.g., "*I have only hearing colleagues at work. Conversations with them won't go beyond simple greetings like 'Hi', 'Bye', etc.*" (P14). Such interactions happen using simple, intuitive signs that hearing colleagues can understand without any sign language training/knowledge. Even offices with interpreters, access to interpreters was limited to meetings. Interpreters were not available for informal, unscheduled water-cooler conversations. This resulted in minimal social interaction for the DHH employees.

Results to the survey question, "*How do you communicate with your colleagues at work?*", show that WhatsApp (81.7%) is the most common app used to communicate with colleagues, followed by email (66.4%) and various video conferencing tools, such as Zoom (48.1%), Microsoft Teams (25.2%) and Google Meet (23.7%). While Zoom is the most commonly used video conferencing tool among our survey respondents, this may only reflect their organization's adoption of specific tools and not necessarily the accessibility of these tools. Coincidentally however, our interview participants preferred Zoom (even in offices using Teams/Meet), as Zoom allows users to quickly shift between participant video tiles on a call (even on a mobile device), thus enabling DHH users to focus on the person speaking or signing. Google Meet on the other hand, automatically identifies the speaker and prominently displays the speaker's video on every user's screen by default, which our participants found "*unusable, as an interpreter was also on the call*". This was also reported as a challenge when viewing shared content such as slide decks which also occupy majority of the screen real-estate by default. Finally, this accessibility challenge also manifests in recordings of meetings where the content recorded depends on automatic choices made by the tools which are not necessarily informed by accessibility for the DHH community.

4.2 Other Everyday Settings

While workplace challenges are the most pressing, our survey results show that the DHH community face communication challenges in a variety of everyday scenarios (Figure 1), which we discuss here.

4.2.1 Phone Calls. Our participants never make audio calls. However, they occasionally receive audio calls. These calls are generally from food delivery personnel (e.g., Swiggy, Zomato), e-commerce delivery personnel (e.g., Amazon, Flipkart), or ride-hailing drivers (e.g., Uber, Ola). The calls are of two types—*confirmatory calls* informing the service requester that they have arrived at their doorstep, and *query calls* asking for guidance to reach the exact address of the service requester. Addresses in India are not well-defined, hence such query calls are not uncommon [18].

> "When I book a cab, it is difficult to guide the person to my place... If I go to the doctor for fever or any problem, I can at least write and show them. But with drivers, there is no way to make them understand. Sometimes I text, but they don't reply and keep calling." – *P9*

The query calls cause distress to our participants as they can neither understand nor reply to the callers. Five participants reported taking their hearing neighbours' or colleagues' help to answer such calls on their behalf. The COVID-19 pandemic has resulted in a steep rise in the usage of home delivery services, thus exacerbating this accessibility problem.

4.2.2 Banks. Bank officials usually do not understand sign language. Our participants complained that hiring an interpreter to accompany them while visiting a bank is prohibitively expensive. Their only option was to take help from a hearing person. However participants (like P8) were hesitant to "*disturb their family members and friends for everyday task*". The combination of lack of interpreters, crowded bank, and complex procedures, makes it a challenging scenario:

> "I went to the bank to change my ATM card. There was already a queue at the counter, so I did not have the time to write and show to him [the bank official] and explain... It was difficult... I went to the bank in Delhi. He said you have to go to Kerala [a state in South India] and do this work. There was a lot of communication problem. He asked me to get someone from my family to assist me. It was my work... why should I bring someone else? Also many people were there, so he can't spend a lot of time with me." – *P13*

Even commonly used workarounds like writing on a piece of paper do not work, due to the time needed for a two-way written conversation. P3 reported bank managers "*getting angry*" at him for the delays caused in writing-based communication, as he was "*holding up the queue*". Despite these difficulties, P10 reported that her DHH school taught her the kind of questions to ask at banks and similar institutions. With this training, she is able to manage independently.

4.2.3 Home. We found that our interview participants faced challenges in communicating with members of their family. In particular, 7 participants stated that their parents do not know sign language, and that conversations with their parents are limited to simple questions and gestures, like "*Did you eat your food?*". A few participants (like P1) are able to lipread their parents. Parents often discourage the use of sign language due to the stigma associated with it and encourage their children to speak. Such social stigma against people with disabilities has been known to result in significant developmental challenges [58]. As communication is crucial for any successful relationship, constrained conversation can negatively impact relationship between the family members. For instance,

> "My parents talk a lot... I taught them a few signs, but they don't like me signing, so they always speak. If I don't understand then they use sign language. I don't have much conversation with them... My parents tell me to practice speaking even if I don't understand it... I don't like to interact with them much." – *P12*

Three participants were able to communicate with their parents in sign language—two of them have Deaf parents and the third taught sign language to his hearing parents.

Our participants reported that their siblings were more receptive to learning sign language. Four participants have hearing siblings who know "*ISL really well*". For conversing with their parents, most participants took the help of their siblings (or a friend) who can both sign and speak. Similar to workplace setting, our survey respondents used WhatsApp most commonly (95.4% of respondents) to interact with family and friends, followed by Zoom (58.0%), Google Meet (30.5%) and Microsoft Teams (23.7%).

4.2.4 Doctor. Our participants found visiting a doctor to be accessible, as they are generally "*patient and educated*", and agree to have a written conversation. However, miscommunication with a doctor can be potentially risky as it can lead to misdiagnosis and/or incorrect medication. Hence, for serious illnesses, we found our participants preferred taking help of a hearing family member, friend, or an interpreter for doctor visits.

> "If it is a small thing like fever then I can write and show. But if I have a major thing, then I have to bring an interpreter for better understanding. I don't know how to simplify the language and explain. It's better to take someone who knows sign language, as it's risky and a big issue if we don't understand what they [the doctor] are saying." – *P13*

Moreover, as per P10, the sentence structure and English grammar used by the DHH community increases the chances of miscommunication in a written conversation.

4.2.5 Transportation. Our participants use three modes of transportation for their daily commute—cabs, auto-rickshaws, and public buses. As discussed in Section 4.2.1, ride-hailing cab drivers call their passengers for confirmation and address-related queries. All our participants have experienced their rides being cancelled by the driver when they did not pick up the driver's phone calls. Attempts by our participants to interact with them over text messages were mostly unsuccessful, as drivers were usually driving when our participants messaged, and hence were unable to check and reply to messages. This inability to communicate with cab drivers emerged as a major challenge, with 9 participants complaining about it. The most common workaround was to take the help of a nearby hearing person to interact with the driver. P4 stated another workaround, wherein his organization provides a free cab pick-up and drop service at a fixed time for its DHH employees. If the employee does not show up on time for his/her pickup, the driver calls the office interpreter and the interpreter checks with the DHH employee.

Despite the phone call related challenges, our participants preferred app-based ride-hailing services over hailing auto-rickshaws on the road: "*It is still much easier to book a cab on the app than to catch an auto [-rickshaw] on the road as it is difficult to explain the destination to auto drivers. In ride-hailing apps, it is easy to put the [destination] address in text.*" (P1). Moreover, due to language differences between auto-rickshaw drivers and our participants, written message-based communication may not be feasible.

Lastly, our participants face unique challenges while using public buses. Public buses are the most affordable means of transportation in India and hence tend to be overcrowded. Moreover, the rider needs to know the correct bus stop to get down at, and the name of the next stop is usually announced by the bus conductor. Thus, a bus ride requires communication with the bus conductor or co-passengers, which is difficult using written messages, both due to the crowd and language differences between co-passengers and our participants. For instance:

> "While travelling in the bus when I ask 'Which place is this?', they [co-passengers] cannot understand and sometimes ignore me. When I write and ask them, people don't understand English because they are from village. That's why I have to write in Tamil and show them the paper." – *P7*

4.2.6 Shops. We inquired about two types of in-person shopping—*frequent purchases* such as buying groceries, and *occasional purchases* such as shopping for clothes. Our participants preferred visiting large supermarkets over small grocery stores, as these supermarkets "*permit self-service*", with prices listed on all products, and aisles clearly labelled with their product categories. In contrast, for shopping clothes, our participants preferred local smaller cloth markets as they are cheaper. (These cloth markets have multiple shops selling similar items.) Participants were comfortable shopping in such markets as their interactions were limited to asking for the price. Moreover, the presence of multiple shops meant that if a shopkeeper was unable to communicate well with them, they could move to the next shop. For instance:

> "I never used to go shopping as my mother always bought clothes for me... After coming to Hyderabad [where I live alone], I started going shopping with [DHH] friends. Now, I buy my own dresses. If I do this sign *[does the sign for money]*, the shopkeeper will hold up fingers to convey the price of the product. If the price is high, I bargain. I will try to write down my price and negotiate. If they don't understand, I will just leave and go the next shop. I don't face any problems." – *P15*

4.3 Insights into Indian Sign Language

In this subsection, we present findings about ISL that we learnt over the course of our study. Specifically, we examine the state of sign language education in India, the influence of ISL on the way the DHH community perceives English, and the difficulties in communication due to the diversity of ISL.

4.3.1 Sign Language Education. A majority of our participants (11) attended schools for DHH children; the remaining 4 participants attended mainstream schools and struggled throughout. Participants mainly attended mainstream schools because their parents did not want them to be seen as "*different*". They recalled that they were given a hearing aid at school and asked to sit in the front, in an attempt to ensure that they could hear and lipread the teachers.

> "I studied with other hearing students near my house. I went to NISH for college where I learnt sign language. It was difficult for me while studying in school. I sat in the first row. I used hearing aids and could understand slightly. Sometimes I could not understand some words and kept asking the teacher. If it gets cold like during the rainy season, then my hearing reduces and I could not hear anything at all." – *P14*

This approach resulted in negative experiences—P2 and P12 also mentioned not able to understand their teachers and relied solely on notes written by their friends, while P14 faced social isolation in school due to his disability.

Interestingly, even participants who attended schools for DHH children were forbidden to use sign language at school, and students were encouraged to lipread and speak instead. P10 mentioned:

> "I can read lips a little now, but I mostly use ISL. I don't speak at all now... When I was a child, the teacher used to teach me and force me to speak. I went to a deaf school where there was a speech therapist, but I could not understand him." – *P10*

A key finding from our survey results is that this encouragement to speak is widely prevalent in India, as 49.6% of our survey respondents responded 'yes' to the question *"Have you had speech therapy?"*. P4 shared that he was encouraged to "*practice speaking slowly and clearly*" by his parents, as they feared that he would be socially isolated if he relied only

on sign language. Despite this, there was an overwhelming consensus amongst our interview participants that they preferred sign language over speech, and were more comfortable using ISL.

Five of our participants did not learn ISL at home or school, but picked it up informally from peers, by watching YouTube videos, or through apps like the DEF-ISL app[3]. Popular YouTube channels mentioned by our survey respondents include ISH News, MBM Vadodra, NAD, The Daily Moth, Yunikee, ISLRTC, and NIOS. Interview participants stated that they found peer learning to be the the most effective way to learn ISL. Three participants mentioned clearing doubts by asking their friends, and stated, "*practising with friends gave the confidence needed to use sign language in public*". All the five participants later learnt ISL formally, either in college as part of their undergraduate studies or by completing a sign language diploma course.

4.3.2 English literacy. As mentioned earlier, using ISL to communicate with the hearing community is often not feasible due to the lack of interpreters. Moreover, these interpreters are very expensive (costing 13.7–20.5 USD/hour). Thus, written English becomes the dominant method to communicate with the hearing people. However, our participants face various communication challenges in writing and reading English, mainly due to linguistic differences between ISL and English. Below are a few examples of the unique sentence structure used by our participants, provided by our interpreter from her WhatsApp conversations with them:

> "I go start my exercise walking now it" instead of "I shall now start my walking exercise".
> "Hard exercise some same gym" instead of "I do some hard exercise in the same gym".
> "Before I wash shirt and jeans" instead of "I washed my shirt and jeans before this".

The first and perhaps most apparent difference between ISL and English is the underlying sentence structure. While English follows the SVO (subject - verb - object) order for arranging words in a sentence, ISL follows the SOV (subject - object - verb) order [79]. Due to this, the DHH community tends to follow the SOV order even while communicating in English, resulting in sentences that are confusing to read. The other major difference between ISL and English is the absence of articles (like 'a' or 'the') and connectives (like 'and' or 'or') in ISL [17]. This results in missing articles/connectives when DHH people write in English too. Finally, P13 (a teacher) noted that DHH students often have difficulty in understanding idioms such as "*time flies*".

4.3.3 Diversity of ISL. ISL is not a single, standardised sign language. Variations in ISL became a frequent topic of conversation during our interviews as our interpreter (who is from Hyderabad) would sometimes fail to understand signs used by our participants who were from different parts of India. For instance:

> "The sign for 'marriage' is shown by a mangalsutra (an auspicious necklace used in Hindu weddings) *[performs a sign that draws out a necklace]* in Chennai, while in Hyderabad it is shown by the holding of hands *[touches right hand fingers with left]*... just like when the bride and groom hold hands during weddings... Similarly in North [India], the sign for mother is *[touches her nose]* which indicates the nose ring, while in Hyderabad you need to do two signs for mother, the North Indian sign followed by this *[holds fist to cheek]*." – *Interpreter*

We observed that these variations in sign language are reflective of India's cultural and linguistic diversity. India is comprised of 29 culturally-diverse states and is home to 184 languages (spoken by more than 10,000 speakers) [41]. ISL, just like spoken languages, is strongly influenced by the culture of its signers. Three of our participants mentioned

[3]https://play.google.com/store/apps/details?id=in.eightfolds.deafenabled

knowing multiple signs for a given word, as they had grown up in multiple cities. P9 stated that even signs for basic words like the days of the week varied across different regions she grew up in.

Participants reported instances where these variations in ISL caused problems during lectures, training workshops, and workplace meetings.

> "Sometimes she [my manager at work who knows sign language] will stop and say 'I can't understand what you are signing.' She'll ask me twice or thrice again. She won't understand as she is from North [India]. Sometimes I get fed up and just text her to explain." – *P1*.

Apart from the ISL diversity, four participants (from tier-1 cities like Delhi and Mumbai) reported mixing of American Sign Language (ASL) with ISL. This phenomenon is similar to code-switching, wherein a speaker alternates between two or more languages in the same conversation or utterance [14]. In particular, we found the ASL-ISL switch similar to the Hindi-English switch (called *Hinglish*), which is commonly observed among Hindi-speaking Indians [59]. Our participants specified that the reason for code-switching is the unique advantage offered by ASL. Unlike the two-handed ISL, all ASL alphabets can be signed using one hand [27], thus enabling them to sign while holding objects with the other hand. All the four participants mentioned using ASL mainly for WhatsApp video calls as they could hold their phone with one hand and use the other hand for signing. However, all of them reported that they were not fluent in ASL, and could only fingerspell the alphabets and know of a few basic ASL signs. Other participants who did not use ASL would sign by using their body or face as a substitute for one hand while holding objects.

5 DISCUSSION

Our survey and interviews highlight key challenges faced by the Indian DHH community on a day-to-day basis. Here, we examine key findings in the broader context of existing work and provide design and policy recommendations.

5.1 Agency

In an effort to gain agency and avoid stigma, our participants were encouraged by their teachers and parents from early childhood to speak, lip-read, use hearing aids, take speech therapy, and avoid signing in public. These methods were prevalent in both mainstream schools and schools for DHH children. While such approaches worked with moderate success for two hard-of-hearing participants, all profoundly deaf participants were unable to and/or uncomfortable with using these methods. Thus, well-meaning efforts by members of the hearing community to increase the agency of DHH individuals actually have the opposite effect. The role of stigma in influencing decision-making amongst the guardians of children with disabilities has been previously reported, particularly in the Indian blind population [47]. They reported parents did not provide white canes to their children with vision impairments mainly to avoid stigma and make their children '*look less blind*' [47], which curtailed their children's physical and mental growth. This suggests that social stigma supersedes agency in India and serves as an additional hurdle for people with disabilities.

We found that the agency of a DHH individual varies across settings. A few factors influencing agency are the nature of interaction, the time available for communication, and the English comprehension ability of the hearing person. For activities like shopping and ordering food at restaurants, we found our participants independently interacting with the hearing person using a combination of basic signs and written messages, as the required interaction was short and there was time available. Our participants faced difficulty in independently communicating with drivers, bank officials, and doctors, as drivers and bank officials usually did not have time to have a written conversation, and conversations with a doctor have a high risk if misinterpreted. In such settings, our participants took the help of an interpreter or a

hearing person (friend, family, colleague, or stranger). However, this reduces their agency and raises privacy concerns as well. Privacy may not be a major concern for short transactional conversations (such as with a driver or delivery personnel), but privacy can be a barrier for critical conversations, like discussing salary hike with a manager at work or while interacting with a doctor about a stigmatized health condition.

5.2 Designing Inclusive Technology

Interpreters on Demand: Access to an interpreter emerged as the most reliable solution for the DHH community. All of our participants mentioned the ease of interacting with hearing individuals in the presence of an interpreter, and mentioned a reduction in anxiety during such conversations. However, accessing interpreters at all times is infeasible due to the high cost of hiring an interpreter and the dearth of certified interpreters in India. Technologies like on-demand interpreter[4] has potential to solve the problem. Using such solutions has multiple advantages—it is cheaper, enables an interpreter to serve more people (as they can switch calls), and allows DHH users affordable on-demand hiring. While similar solutions are available in India[5], none of our participants were aware of it. We identify a strong requirement for on-demand interpreters in a variety of settings, and believe that such an app would be potentially useful if effectively advertised to the Indian DHH community.

Accessible Video Conferencing Platforms: Video conferencing was used often by our participants, mainly to attend office meetings. Usage of video conferencing platforms increased significantly due to COVID-19. Most of the available platforms (like Zoom, Google Meet and Microsoft Teams) support speech-to-text (STT) technology to auto-generate live captions, serving as a partial technological alternative to a human interpreter. Though our participants found live captions useful, they were unable to read the generated captions at the required pace due to their limited English proficiency. A simple solution is to allow a history of the generated captions which the DHH individual can revisit. A more technically challenging solution would be to integrate automatic tools for text simplification [77] such as Lexi [19]. Another challenge with video conferencing tools is the design of the screen layout. Tools like Google Meet default the primary video on each user's screen to the current speaker (based on audio signals) or the shared content (such as slide deck). This design choice makes the tool inaccessible to the DHH community, who would like to see the interpreter prominently at all times. To address this challenge, video conferencing tools should provide more agency to DHH users by allowing manual configurations of the on-screen layout.

Accessible Audio Phone Calls: Our participants pointed challenges in attending audio phone calls. Recent smartphone apps (such as Rogervoice[6]) attempt to make audio phone calls accessible for the DHH community by extensively using STT and TTS (text-to-speech) technologies. Rogervoice auto-generate captions for incoming audio calls, and provides an optional pre-recorded message to inform the caller that they are speaking to a DHH person. The DHH person can either talk to the caller or type out a message that is read to the caller using TTS technology. Unfortunately, no similar app is available in India. We acknowledge that the technological barriers to such an app are higher in India, given the diversity in languages and accents, and presence of code-mixing.

Ambient Conversation: Our participants struggled to participate in informal social conversations with their colleagues. Similar challenge was reported among DHH employees in the US [37]. Even listening to ongoing conversations while not contributing can help DHH employees connect with their colleagues. Smartphone apps (like Talk to Deaf[7]) enable DHH users to listen to ambient conversations, by using STT technology to display captions for conversations happening

[4]Jeenie: https://jeenie.com/
[5]SignAble: https://play.google.com/store/apps/details?id=org.signable.apprtc
[6]Rogervoice: https://play.google.com/store/apps/details?id=com.rogervoice.app
[7]https://play.google.com/store/apps/details?id=unique2040.com.text2speech

around the user's smartphone. However, we found that these apps worked only when a person is speaking directly into the smartphone's microphone in environments with minimal ambient noise, thus severely limiting their utility. Improved microphone technology for indoor settings is available on consumer-grade smart speaker devices, which should add support to help the DHH listen to ambient conversations.

Accessible Service Apps: While efforts to build novel applications to improve accessibility continues, several problems faced by the DHH community can be solved by designing existing applications more inclusively. In particular, a majority of our participants faced challenges in interacting with cab drivers and e-commerce/food delivery personnel over phone calls, and have to seek help from family, friends, neighbors, colleagues, or strangers. These service apps should add a "Do not disturb mode" or "DHH mode" informing the driver/delivery personnel that the service requester is a DHH individual. In such cases, communication should be automatically restricted to text messages, instead of phone calls.

"Broken" English: The underlying difference in sentence structure between English and ISL, and the limited English proficiency of our participants resulted in incomprehensible English sentences. A similar phenomenon has been observed in other countries as well. For instance, Italian Sign Language has an underlying SOV structure whereas the Italian language follows the SVO order, which results in non-native signers of Italian Sign Language following the SVO order while signing [26]. A tool that inputs text in ISL-like SOV structure and generates text in SVO structure as output with appropriate grammar checks may address this accessibility gap. Moreover, the output text can be translated to local languages using existing Neural Machine Translators, potentially resolving challenges faced by the DHH community as reported while using public buses for transportation.

Conversion between ISL and text: Automatically converting sign language videos, as captured by a smartphone camera, to natural language text is a challenging task. However, this is an active area of research in the AI community [21, 29, 80]. Like other AI tools, training a sign language recognition system requires large amounts of training data, consisting of sign videos and their corresponding text translations. Given its diversity, a large and concerted effort is required to create such datasets for ISL. Further, such data collection effort may initially be focused on the most critical settings, such as communication at the workplace, and with doctors, bank officials and service providers. Integrating such a tool would be most useful in communication apps like WhatsApp, which is used by 95.7% of survey respondents for personal calls and 81.7% of them for professional calls. This would require that the sign language recognition tool should support low-to-medium end smartphones as well.

5.3 Policy Recommendations

While technological solutions can help address some of the identified challenges, many others are structural in nature and require policy intervention. For example, the taboo associated with using sign language in schools was exacerbated by the lack of official recognition of ISL as a language subject. This was recently remedied in July 2021 by the National Education Policy [11] that gave ISL the status of a language subject. The policy also calls for the standardisation of ISL while allowing regional variants to be taught, thereby alleviating some of the problems that arise due to ISL's diversity. Another laudable initiative on improving accessibility in schools is the Haryana (a state in North India) government's translation of state school textbooks to Indian Sign Language [43]. India national level board of education, Central Board of Secondary Education, should consider releasing similar translations for all textbooks.

Moreover, to solve the lack of interpreters problem, policy intervention and training is required to increase the number of interpreters and guarantee access to them for DHH individuals. Mandates to generate accessible content, such as the Ministry of Information & Broadcasting's memorandum to deliver subtitles/closed captioning/sign language options across news television programs [10] is also a welcome measure. As noted before, steps should be taken to

ensure that the language used in these captions is simple enough to be understood by the DHH community. Another influential step towards ensuring access to interpreters and content in ISL would be recognizing ISL as an official language of India, which would encourage its development, preservation, and enrichment.

5.4 Limitations

This study provides a rich description of the lives of the employed DHH community in India. There are a few limitations to this study which the reader should consider when determining how these findings may generalize to other contexts. First, while the introduction to the survey was circulated in ISL, the survey was only available in English. This limited the audience of the survey to DHH who could read and write in English. Since the survey was targeted at the employed DHH community, it is reasonable to assume English literacy amongst our target respondents. Second, due to the diversity in ISL, our interpreter had difficulty understanding a few participants during our interviews. The issue was mitigated as much as possible by fingerspelling and repeatedly clarifying what the participant was saying. Third, the small sample size limited our analyses. A larger number of participants is required to identify broader trends.

6 CONCLUSION

In this work, we study the challenges faced by the employed DHH community in India, by interviewing 15 DHH participants and surveying 131 DHH respondents. We emphasize the specificity of our study here as a reminder to readers that this study is at best a first step towards characterizing accessibility challenges for the DHH community outside of the developed regions context. Our study reveals various challenges faced by the employed Indian DHH community on a day-to-day basis, in a variety of settings including workplace, home, commute, and bank. We also discuss foundational challenges due to the stigma associated with signing in India, the English literacy issues faced by the DHH community, and the problems arising due to the variations in ISL, which is characteristically different from other sign languages like ASL. Lastly, we highlight technology-related challenges, where we highlight problems with video calling applications, automated captioning services, and app-based service delivery. We cby contrasting the Indian workplace challenges with the experiences of DHH employees in the Global North, and propose technology and design recommendations to tackle the three types of identified challenges.

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